

The Angenent Oval under Curve Shortening: Exact Ancient Geometry and Two Grim-Reaper Ends

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Abstract

For the central component of $\cos x = e^t \cosh y$ in $|x| < \pi/2$, $t < 0$, we give a convention-explicit proof of curve shortening, smooth embedded strict convexity, and exact curvature extrema. We close its width, height, area and elliptic length, identify the exact arrival-time foliation of the punctured strip, and prove both round extinction and the translated upper/lower Grim-Reaper limits.

1 Exact ancient solution and curvature

We use the convention that a convex curve has positive curvature κ with respect to its outward normal n , and curve shortening has outward normal velocity $V_n = -\kappa$. Put $r = e^t \in (0, 1)$ and

$$F(x, y, t) = \cos x - r \cosh y, \quad \Gamma_t = \{F = 0, |x| < \pi/2\}.$$

Theorem 1 (Core implicit solution). *For every $t < 0$, the central component Γ_t is a smooth embedded strictly convex closed curve. It evolves by $V_n = -\kappa$, where*

$$\kappa(x, y, t) = \frac{\cos x}{\sqrt{1 - e^{2t}}}, \quad \frac{e^t}{\sqrt{1 - e^{2t}}} \leq \kappa \leq \frac{1}{\sqrt{1 - e^{2t}}}. \quad (1)$$

It is ancient on $(-\infty, 0)$ and contracts to the origin as $t \uparrow 0$.

Proof. The central component consists of the two graphs

$$y = \pm \operatorname{arcosh}(r^{-1} \cos x), \quad |x| \leq \arccos r.$$

On this level,

$$|\nabla F|^2 = \sin^2 x + r^2 \sinh^2 y = 1 - r^2 > 0, \quad D^2 F = -\cos x \, I. \quad (2)$$

Thus the level is smooth and compact. For $n = -\nabla F/|\nabla F|$, the implicit curvature formula and (2) give (1), which is positive because $r \leq \cos x \leq 1$. Differentiating $F = 0$ along a moving point yields $V_n = F_t/|\nabla F| = -\cos x/\sqrt{1 - r^2} = -\kappa$. The displayed graphs shrink to $(0, 0)$ as $r \uparrow 1$ and exist for every $r > 0$, proving the temporal statements and the sharp extrema. The unrestricted zero set of F is the disjoint union $\bigsqcup_{k \in \mathbb{Z}} (\Gamma_t + (2\pi k, 0))$; it is not one compact curve. $\square$

2 Global geometry, arrival time, and two asymptotic regimes

Write $K(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \phi)^{-1/2} d\phi$; here k is the elliptic *modulus*.

Theorem 2 (Complete geometry and limits). *The horizontal width W , vertical height H , enclosed area A and length L are*

$$W(t) = 2 \arccos(e^t), \quad H(t) = 2 \operatorname{arcosh}(e^{-t}), \quad (3)$$

$$A(t) = -2\pi t, \quad L(t) = 4\sqrt{1 - e^{2t}} K\left(\sqrt{1 - e^{2t}}\right). \quad (4)$$

The negative-time curves foliate $(-\pi/2, \pi/2) \times \mathbb{R} \setminus \{(0, 0)\}$ exactly once. Their arrival time

$$T(x, y) = \log \cos x - \log \cosh y \quad (5)$$

satisfies, away from the origin,

$$-|\nabla T| \operatorname{div} \left(\frac{\nabla T}{|\nabla T|} \right) = 1. \quad (6)$$

Moreover,

$$(-2t)^{-1/2} \Gamma_t \longrightarrow S^1 \quad \text{smoothly as } t \uparrow 0.$$

If $y_+(x, t) = \operatorname{arcosh}(e^{-t} \cos x)$ and $h(t) = \operatorname{arcosh}(e^{-t})$, then

$$y_+(x, t) - h(t) \longrightarrow \log \cos x \quad (7)$$

smoothly on compact subsets of $(-\pi/2, \pi/2)$ as $t \rightarrow -\infty$; the lower tip has the reflected limit.

Proof. The extrema of the two graphs give (3). On the upper graph,

$$y_x = -\frac{\sin x}{\sqrt{\cos^2 x - r^2}}, \quad ds = \frac{\sqrt{1 - r^2}}{\sqrt{\cos^2 x - r^2}} dx.$$

The substitution $\sin x = \sqrt{1 - r^2} \sin \phi$, followed by the two graph and two half-curve symmetries, proves the length formula. Directly differentiating the area integral (the endpoint value is zero) gives

$$A'(t) = -4 \int_0^{\arccos r} \frac{\cos x}{\sqrt{\cos^2 x - r^2}} dx = -2\pi.$$

Since $A(t) \rightarrow 0$ as $t \uparrow 0$, the area formula follows.

On the open strip, $T \leq 0$, with equality only at the origin, and $T = t$ is equivalent to the oval equation. This proves the precise punctured-strip foliation. Setting $a = -\tan x$, $b = -\tanh y$ in the divergence of $(a, b)/\sqrt{a^2 + b^2}$ reduces the left side of (6) to

$$\frac{\sec^2 x \tanh^2 y + \operatorname{sech}^2 y \tan^2 x}{\tan^2 x + \tanh^2 y} = 1.$$

In tangent-angle coordinates (up to an angle shift),

$$\kappa^2(\theta, t) = \frac{1}{1 - e^{2t}} - \sin^2 \theta. \quad (8)$$

After scaling, $(-2t)\kappa^2 \rightarrow 1$ with every angular derivative. Symmetry fixes translation, so the fundamental theorem for strictly convex plane curves gives the smooth circular limit. Finally, $\operatorname{arcosh} z = \log(2z) + O(z^{-2})$ with differentiated uniformity away from the strip edges. Apply it to $z = e^{-t} \cos x$ and to $z = e^{-t}$ to obtain (7); reflection gives the lower statement. $\square$

Source lineage

References

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- [3] T. Bourni, M. Langford, and G. Tinaglia, “Convex ancient solutions to curve shortening flow,” arXiv:[1903.02022](https://arxiv.org/abs/1903.02022).